

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE							
Regular/Supplementary Winter Examination - 2024							
Course: B.Tech			Branch : Common To All Branches				
Semester : I							
Subject Code & Name: BTES104 Computer Programming in C							
Max Marks: 60			Date: 13/02/2025				
Duration: 3 Hr.							
Instructions to the Students: 1. Each question carries 12 marks. 2. Question No. 1 will be compulsory and include objective-type questions. 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6. 4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 5. Use of non-programmable scientific calculators is allowed. 6. Assume suitable data wherever necessary and mention it clearly.							
						(Level/CO)	Mark s
Q.1	Objective type questions. (Compulsory Question)						12
	What is the primary function of a compiler in programming?					CO1	1
	a. Execute the program	b. Convert source code to machine code	c. Debug the program	d. Write source code			
2	What does a flowchart primarily represent?					CO1	1
	a. Hardware connections	b. Algorithm or process flow	c. Database relationships	d. User interface design			
3	What is the output of the expression 5 % 2 in C?					CO2	1
	a. 0	b. 2	c. 1	d. 5			
4	What is the output of the expression 10 > 5 && 3 < 2?					CO2	1
	a. 0	b. 1	c. True	d. False			
5	Which statement allows skipping the current iteration of a loop and proceeding to the next?					CO3	1
	a. break	b. continue	c. exit	d. return			
6	Which loop ensures execution of the loop body at least once?					CO3	1

	a. while	b. do-while	c. for	d. infinite loop		
7	How are arrays indexed in C?				CO4	1
	a. 0-based	b. 1-based	c. -1-based	d. Custom index		
8	What is the maximum number of elements in an array int arr[10][20];?				CO4	1
	a. 10	b. 20	c. 200	d. 100		
9	What does the increment operator (++) do?				CO2	1
	a. Multiplies a variable by 2	b. Increments a variable by 1	c. Adds 2 to a variable	d. Divides a variable by 2		
10	Which control flow statement is used for multi-way branching in C?				CO3	1
	a. if-else	b. switch	c. while	d. do-while		
11	Which keyword is used to define a structure in C?				CO5	1
	a. struct	b. class	c. object	d. record		
12	What is a pointer in C?				CO5	1
	a. A variable that stores the address of another variable	b. A variable that stores multiple values	c. A function returning integers	d. A structure type		
Q.2	Solve the following.					12
A)	Explain different phases in programming process.				CO1	6
B)	Draw the flowchart, write algorithm to check the given age is valid for voting.				CO1	6
Q.3	Solve the following.					12
A)	Write short notes on logical operators with examples.				CO2	6
B)	Explain the different Data types in C with suitable example.				CO2	6

Q. 4	Solve Any Two of the following.		12
A)	Write the syntax, flow chart and explain with example. i) for loop ii) while loop iii) do while	CO3	6
B)	Write a program in C to swap two numbers using a function.	CO3	6
C)	Write a program to check the given number is positive, negative or zero.	CO3	6
Q.5	Solve Any Two of the following.		12
A)	Write a program in C to read n number of values in an array and display them in reverse order.	CO4	6
B)	Write the any three built-in functions in string with description and syntax.	CO4	6
C)	Write a program in C to separate odd and even integers into separate arrays.	CO4	6
Q.6	Solve Any Two of the following.		12
A)	Explain structures with example.	CO5	6
B)	Differentiate between array and structure with suitable example.	CO5	6
C)	Write a C program to store a person's information(name, Empno and Salary) and display it on the screen using structure.	CO5	6
*** End ***			